Ayush Negi

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Profile

A Software Developer Engineer - currently with more than 1.6 years of building backend systems for a national-scale government cybersecurity platform, with hands-on AI experience in LLMs, RAG and multi-agent systems. Co-authored a published research paper and now applying for an MSc in Artificial Intelligence to move from applying AI tools to understanding and evaluating why they work.

Work Experience

Software Developer Apr 2025 – Present

Zindagi Technologies *New Delhi, India*

Building the Government of India's National Cyber Range Platform

- Architected scalable REST APIs using **FastAPI** with object-oriented design across distributed, multi-tiered systems handling national-scale cybersecurity workflows.
- Reduced API latency **30–40%** via **PostgreSQL** redesign; built event-driven microservices with **Kafka/RabbitMQ**, cutting inter-service latency **35%**.
- Owned **CI/CD** (GitLab) and **Kubernetes** deployments via Helm, including cluster monitoring.
- Conducted code reviews, enforced Agile/Scrum practices, and mentored **5+ interns**; collaborated with product and security teams on regulatory compliance.

Software Developer Trainee Feb 2025 – Mar 2025

National Informatics Centre – Govt. of India *New Delhi, India*

- Modernized a legacy government database using **Django** & RESTful APIs, cutting query response times **60%**.
- Optimized data retrieval with **MongoDB** and **Redis** caching; automated backup and recovery.

Software Developer Intern Jul 2024 – Aug 2024

DRDO – Ministry of Defence, Govt. of India *New Delhi, India*

- Built a real-time **computer vision** surveillance system using **CNNs**, **OpenCV**, **LLMs** and **object detection** in **PyTorch**, covering preprocessing, model inference and performance optimization.
- **Atomic Clock Drift Simulation** (Python, ANN, NumPy): modeled cesium-133 clock drift with synthetic noise and trained an **ANN** to predict and correct it, reaching ∼**99.99%** accuracy on the simulated data.

Key Projects

More to find on portfolio

MR-Review-Agent | *Multi-Agent LLM, RAG, LangGraph, Python* GitHub

- Designed a **multi-agent PR review system**: an orchestrator sends a code diff to specialized **security, logic and performance agents**, and a **critic agent** synthesizes their findings.
- Cut agent-stage latency ∼**58%** (est. 32s → ∼13.5s) by moving from sequential to **concurrent** execution, verified with per-agent timing instrumentation.
- Traced and fixed a silent **RAG retrieval bug** (query embedding wrapped in an extra list layer) that broke the security agent's context step; now refactoring orchestration to **LangGraph** for explicit state and failure handling.

Agentic-AI: Production RAG Chatbot | *RAG, RAGAS, Python* GitHub

- Built and maintain a production **Retrieval-Augmented Generation** chatbot.
- Instrumenting it with **RAGAS** (faithfulness, answer relevance, context precision/recall) to replace informal spot-checks with quantitative evaluation of retrieval and chunking choices.

Heart Disease Prediction | *Python, Flask, scikit-learn, Pandas, JavaScript* GitHub

- Built a **Flask web app** and ML model that predicts heart disease from patient inputs (age, gender, chest pain type, cholesterol, etc.); users can print a report to track their condition. Motivated by CVDs causing 32% of global deaths (WHO, 2019).
- Trained and compared **Logistic Regression, SVM, Decision Tree, Random Forest and KNN**; data work in NumPy, Pandas, Matplotlib.

Technical Skills

Languages: Python, JavaScript, SQL, C, HTML, CSS
AI & ML: RAG, Multi-Agent Systems, LangGraph, LLMs, RAGAS Evaluation, Deep Learning, CNNs, ANNs, Computer Vision, OpenCV, PyTorch.
Backend & API Design: FastAPI, Flask, Django, Pydantic, RESTful APIs, Microservices, Concurrency
Distributed Systems: Apache Kafka, RabbitMQ, Event-Driven Architecture

Databases & Caching: PostgreSQL, Oracle, MongoDB, Redis, InfluxDB
Cloud & DevOps: OpenStack, AWS, MinIO, Docker, Kubernetes, Helm, Linux, GitLab CI/CD, Git, Prometheus, Grafana
Testing & Practices: PyTest, Postman, Swagger/OpenAPI, Agile, Scrum, Code Review
Tools: VS Code, PyCharm, Jupyter Notebook, K9s

Certifications

Verify on portfolio / LinkedIn

- **IBM SkillsBuild** – Intelligent by Design: Build an AI Agent
- **Udemy** – Complete Python With DSA Bootcamp + LEETCODE Exercises
- **Coding Blocks** – Data Science Master Course

Education

Inderprastha Engineering College, AKTU — Ghaziabad, India
B.Tech – Computer Science Engineering (Ist division) — *2021 – 2025*

Hans Raj Smarak Sr. Sec. School — New Delhi, India
CBSE Class XII – Non Medical (Ist division) — *2020 – 2021*

Achievements & Recognition

- **NCX 2025 (National Cyber Exercise)** – Invited as Zindagi Technologies' developer representative at India's premier national-level cyber defense simulation. [View Post]
- **Research Publication** – Published an academic paper on Blockchain Technology. [View Paper]